Piyush Chaudhary

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SUMMARY

Final-year B.Tech Computer Engineering student who builds document-processing, cloud and mobile software. Currently an intern at SBI Global IT Centre, previously an intern at Mazagon Dock Shipbuilders Limited.

EDUCATION

Bachelor of Technology, Computer Engineering, K.J. Somaiya College of Engineering Final year
Coursework: Natural Language Processing, Cloud Computing, AR/VR, Network and Information Security

EXPERIENCE

Intern, State Bank of India, Global IT Centre Present

- Building a production pipeline that extracts structured tables from scanned telecom audit PDFs into Excel, using a hybrid of computer vision (PyMuPDF, OpenCV) and a vision-language model.
- Compared three architectures on the same documents (Table Transformer, a pure vision-language model, and Docling) before choosing the hybrid design.
- Added a fast pre-scan that locates key header fields and the conclusion-table page range before the per-page pass, and switched to lazy page generators after eager rendering exhausted memory.
- Fixed a column-splicing bug that turned an 11-column table into 24 phantom columns, and added a guard that rejects implausible vision-language model cell text.
- Generated synthetic scanned PDFs with noise and skew for testing, and documented the system in a 16-page technical report written in LaTeX.

Intern, Mazagon Dock Shipbuilders Limited

PROJECTS

The Tangled Path: Choice-Based Anti-Drug Game (HTML, CSS, JavaScript) sudo-choices.vercel.app

- 1st runner up at a hackathon. Team leader and front-end developer on a team of four.
- A story game that follows one character through four life stages (ages 6, 15, 25 and 70), where each choice leads toward a good or a bad ending.
- Includes a built-in story teller and a full-screen mode. Deployed on Vercel.

MouseShare (Python, Kotlin, Android)

- Shares a Windows PC's mouse and keyboard with an Android phone: a Python desktop app captures input and a Kotlin Android app receives and acts on it.
- Comparing transports (HID keyboard and mouse, WiFi) to cut input lag.

GIF Crop and Overlay Tool (JavaScript, HTML, Canvas) piyushchy.github.io/gif-tool

- Browser tool that crops a GIF to a freehand or polygon shape and overlays one GIF on another with per-frame controls (blank, skip, hold, and shifting one track against the other).
- Wrote the GIF decoder and encoder from scratch (LZW and median-cut palette). Runs entirely in the browser, so nothing is uploaded.

Cloud-Native App on AWS (Docker, AWS ECR, ECS Fargate, ALB, RDS MySQL, CodePipeline, SNS)

- Containerized a Node.js app and deployed it on ECS Fargate behind an Application Load Balancer, with the image stored in ECR.
- Connected to an RDS MySQL database through a bastion host over Session Manager, and added a CodePipeline CI/CD flow and an SNS alarm.

RNN vs LSTM Text Classification (Python, PyTorch, Google Colab)

- Trained and compared a plain RNN and an LSTM on the same text classification task in a runnable Google Colab notebook.

TECHNICAL SKILLS

Languages: Python, Kotlin, JavaScript, HTML, CSS, LaTeX
Computer vision and document AI: OpenCV, PyMuPDF, EasyOCR, Table Transformer, Docling, Pillow, vision-language models
Machine learning: PyTorch, RNN, LSTM
Cloud and infrastructure: Docker, AWS (ECR, ECS Fargate, ALB, RDS MySQL, EC2, CodePipeline, SNS)
Tools: Git, GitHub, Google Colab, openpyxl, Excel automation

ACHIEVEMENTS

- 1st runner up, hackathon, for The Tangled Path (team leader and front-end developer)